LIREAD

Small daily pages. Massive compounding wisdom.

A serene, distraction-free reading sanctuary with deep-focus timers, procedural ambient soundscapes, active in-reader notification blocking, offline reading, and mindful community accountability.

Web & PWA | React 18 + TypeScript | Supabase PostgreSQL | Web Audio API

The Liread Mission

Restoring sustained human contemplation in an era of micro-stimuli

> ## "We are starving for wisdom while drowning in algorithmic noise."
>
> Liread is engineered to reverse attention fragmentation. Rather than competing with social feeds for fleeting dopamine, Liread provides a sacred digital space where readers cultivate uninterrupted focus, silence distractions, cherish classic literature, and build lifelong reading consistency.

The Screen Time Displacement

Algorithmic feeds have quietly displaced reflective reading

144m

DAILY ALGORITHMIC SCREEN TIME

The average adult spends over 2.4 hours every day scrolling short-form video and social media feeds that break continuous attention into 15-second fragments.

< 14m

DAILY DEEP BOOK READING

Time allocated to literary reading has fallen over 40% in ten years. Readers desire to read books, but algorithmic friction continuously sabotages their intentions.

The Abandonment Epidemic

Starting books is easy; sustaining momentum to the last page is rare

74%

OF STARTED BOOKS ARE NEVER FINISHED

Without a structured daily focus timer, gentle milestone pacing, and an environment shielded from notifications, 3 out of 4 books are abandoned around page 60. Notification interruptions break flow and destroy the cognitive bridge required to finish long-form masterworks.

The Solitary Reader Dilemma

Fitness has Strava, coding has GitHub. Reading remains completely isolated.

Why Are Healthy Habits Sustained in Public, but Reading Suffers in Private?

Habit formation research proves that peer accountability increases habit completion rates by over **65%**. Yet readers are forced between two extremes: solitary isolation with zero reinforcement, or noisy commercial book review sites dominated by ads and algorithmic rage-bait.

The Missing Element: A quiet, pressure-free sanctuary that celebrates daily consistency without vanity metrics or toxic competition.

The Lost Marginalia Paradox

Humanity reads brilliant insights, only to let them evaporate

Highlights Locked in Digital Graveyards

When readers encounter a life-defining quote or a philosophical breakthrough, it is either underlined in pencil never to be referenced again, or locked in proprietary e-reader files. There is no fluid commonplace book system to distill, preserve, and exchange timeless passages with fellow thinkers.

✗ The Status Quo

Underlined text forgotten on a physical shelf or lost in proprietary formats.

✓ The Liread Way

A single click transforms book marginalia into personal saved wisdom and community discussion.

Introducing Liread

A mindful digital sanctuary uniting deep focus and literary community

Focus in Private. Inspire in Public.

Liread provides an all-in-one distraction-free web sanctuary: an immersive reading HUD timer, active in-reader notification and app blocking, zero-network procedural audio, browser-native EPUB/PDF reader, and a high-signal social feed built strictly around authentic reflections.

✦ The Reading Sanctuary & HUD
Full-screen timer with customizable daily goals and procedural soundscapes.

✦ In-Reader App & Notification Blocker
Mutes distractions, locks focus, and prevents app-switching during reading.

✦ The Open Library & Offline Reader
Read millions of public domain books offline in your browser with zero latency.

✦ The Commonplace Feed
Publish verified marginalia and celebrate daily consistency with real readers.

The Philosophy of Compounding

Small daily pages compound into monumental lifelong wisdom

18 Books

FROM JUST 15 MINUTES DAILY

Reading at standard cadence for only 15 minutes each morning yields over **1,200,000 words**—equivalent to completing 18 full masterworks every single year.

109 Hours

OF NOTIFICATION-FREE FOCUS

Over 4.5 solid days of continuous cognitive contemplation gained back from algorithm doom-scrolling, strengthening attention and cognitive reserve.

Distraction-Free Reading Timer

A focused HUD designed to anchor uninterrupted reading sessions

Full-Screen Focus Mode with Daily Goal Arcs

Implemented in `FullTimerModal.tsx` and `FloatingTimerBar.tsx`, the timer provides a gentle, ambient stopwatch that keeps time without aggressive beeping.

- **Custom Daily Commitments:** Quick presets (15m, 30m, 45m, 60m) that automatically advance your streak.
- **Floating Sticky Bar:** Keep reading while browsing your library or exploring new genres without losing session time.
- **Automated Database Logging:** Seamlessly writes to `reading_sessions` upon completion.

Procedural Web Audio Engine

Pure mathematical soundscapes with zero network downloads

100% Client-Side Audio Synthesis via Mathematical Oscillators

In `ambientAudioEngine.ts`, Liread bypasses heavy MP3 streams. Using the native browser Web Audio API, oscillators, and biquad noise filters, it synthesizes infinite, non-repeating acoustic environments.



Calm Rain
Pink noise filtering



Fireplace
Stochastic crackles



Deep Space
Sub-bass drone



Morning Birds
Harmonic chirps

ReadEra Offline Reader Engine

In-browser EPUB and PDF reading backed by client IndexedDB storage

Seamless In-Browser Reading with Zero External Software

Powered by `ReadEraViewer.tsx` and `readerStorageService.ts`, Liread turns modern browsers into rich e-readers supporting EPUB reflowable text and PDF rendering.

- **IndexedDB Persistence:** Uploaded books and downloaded classics are stored locally in the `readera-books-db` object store.
- **Offline Resilient:** Read masterworks on airplanes or trains with full page state preservation.
- **Format Adaptability:** Dynamic font scaling, line height tuning, and dark parchment modes.

Deep Immersion Shield: App & Notification Blocker

Active distraction suppression locking focus and silencing digital noise in the e-reader

Hardware-Level Distraction Shielding Directly in the Web Browser

Implemented in `FocusShieldModal.tsx` and `focusShieldStore.ts`, Liread actively eliminates the 23-minute cognitive penalty of micro-interruptions by turning standard browsers into dedicated distraction-free reading terminals.

- **Notification Silencing Shield:** Automatically silences in-app notifications, tab badges, and sound alerts while inside the reader to prevent breaking flow.
- **Anti-Switching App Guard & Visibility Trap:** Detects window blurring, pauses the active reading timer when users navigate away, and triggers a gentle refocus intervention upon return.
- **Screen WakeLock & Kiosk Fullscreen:** Utilizes the Screen WakeLock API to keep displays awake without sleep timeouts, plus fullscreen lock to hide OS taskbars.
- **Accidental Exit Interceptor:** Safeguards reading progress with navigation guards and confirmation dialogs against unintended page closures.

Dynamic Literary Discovery

Real-time Open Library integration across 14 curated genres

Instant Access to Millions of Works without Paywalls

Implemented in `ExploreView.tsx` and `bookSearchService.ts`, Liread queries Open Library's Subjects API and Gutendex with intelligent dual in-memory and `sessionStorage` caching.

- **14 Diverse Curated Shelves:** Philosophy, Technology, Science & Nature, Psychology, Habits & Focus, Classics, Sci-Fi, and History.
- **The Book of the Week Spotlight:** Dynamic algorithmic spotlight highlighting weekly classics with synopsis and quick-start timer.
- **Instant Cover Fallback:** HTML5 Canvas generates authentic clothbound book covers when remote cover images are missing.

Social Marginalia & Commonplace Feed

Sharing passages and reflections with strictly verified human engagement

High-Signal Literary Exchange with Zero Fake Numbers

In `PostCard.tsx` and `feedStore.ts`, readers publish book quotes, ratings, and chapter takeaways. Likes and comments are calculated from actual relational database rows.

- **Honest Zero States:** If a passage has zero likes, it explicitly says `"No likes yet"` instead of generating fake bots.
- **One-Click Passage Stash:** Bookmark passages shared by other readers directly into your private `Saved Passages` commonplace book.
- **Spring Micro-Interactions:** Delightful animated counters and spring-bounce physics upon upvoting.

Live Co-Reading Sanctuary

Quiet digital solidarity: seeing fellow readers reading across the globe in real time

Real-Time WebSocket Presence via Supabase Channels

Engineered in `coReadingService.ts` and `SanctuaryView.tsx`, the sanctuary connects readers without invasive video calls or noisy chat rooms.

- **Presence Broadcast:** When your reading timer starts, your presence joins the global sanctuary heartbeat.
- **Calm Peer Visibility:** See that 14 other readers are currently reading *Meditations* or *Atomic Habits* right now.
- **Gentle Motivation:** Quiet companionship that eliminates the solitary feeling of reading without social fatigue.

Digital Bookshelf & Streak Analytics

Visualizing personal milestones and compounding reading momentum

Reader Progression Tiers from Novice to Master of Tomes

In **`BookshelfView.tsx`** and **`AnalyticsView.tsx`**, readers track reading statuses (`want_to_read`, `reading`, `completed`) and observe their habit compounding.

- **Automated Streak Calculations:** Consecutively logged reading days automatically advance user streaks.
- **Rank Tiers:** Earn authentic rank promotions: *Novice Reader* ➔ *Bookworm* ➔ *Avid Reader* ➔ *Literary Scholar* ➔ *Master of Tomes*.
- **Private Margins Stash:** Distinct separation between public shared reflections and private reading marginalia.

System Architecture Overview

A high-performance modern web stack built for zero latency and offline capability

1. Client Tier

React 18 + TypeScript + Vite. Chunked asset distribution (<125kB gzip), Zustand state management, Web Audio synthesis, IndexedDB storage.

2. Edge & Serverless

Vercel Serverless Functions (`/api/book-file.ts`, `/api/telegram.ts`). Streamed public domain CDN proxying with automatic fallback resolution.

3. Cloud Database

Supabase PostgreSQL 15 with strict Row-Level Security (RLS), Realtime WebSockets, automated triggers, and PL/pgSQL ranking functions.

Client-Side Architecture & Performance

Strict TypeScript validation, sub-second loads, and reactive state stores

Deterministic State Machines with Zustand

Eliminating unnecessary React re-renders through targeted selectors. State stores (**`authStore`**, **`feedStore`**, **`timerStore`**, **`readerStore`**) maintain persistent local state with automatic rehydration sanitizers.

- Zero compiler errors with strict **`tsc -b && vite build`**.
- Vendor chunk splitting isolating PDF.js, Charts, and Supabase client code.

```
// Optimized persistent feed state with auto-sanitization export const useFeedStore = create<FeedStore>()( persist((set, get) => ({ posts: [], sortBy: 'latest', toggleUpvote: async (postId, userId) => { // Optimistic spring UI update set((state) => ({ ... })); await postsService.toggleUpvote(postId, userId); } }), { name: 'liread_feed_state_v2' }) );
```

Serverless Edge & CDN Proxying

Delivering high-speed book streams while overcoming CORS limitations

Self-Contained Edge Proxy (`/api/book-file.ts`)

When readers request timeless works from Project Gutenberg or Internet Archive, browser CORS constraints often block direct downloads. Liread's edge proxy streams the files dynamically with gzip chunking and 24-hour edge cache headers.

- **Dual Repository Fallback:** If an EPUB link fails, the edge handler seamlessly falls back to PDF copies.
- **Zero-Dependency Registry:** Integrated public domain catalog resolving timeless books in 0ms.
- **Telegram Bot Webhook:** `/api/telegram.ts` handles async bot interactions and book fulfillment.

Relational Entity Model

Clean, normalized PostgreSQL schema modeling the literary domain

```
TABLE profiles ( id UUID PK, username TEXT UNIQUE, total_minutes INT, current_streak INT, rank_tier TEXT, daily_goal_minutes INT );
```

```
TABLE posts ( id UUID PK, user_id UUID FK, book_title TEXT, quote_text TEXT, upvotes_count INT, comments_count INT );
```

```
TABLE reading_sessions ( id UUID PK, user_id UUID FK, book_title TEXT, duration_minutes INT, session_date DATE );
```

All models are connected via foreign keys with cascade deletion. Indexes are placed on **(user_id, session_date)** and **(post_id)** for fast query planning.

Data Integrity: Verified Real Likes

Eliminating fake counters via relational constraints and nested querying

Enforcing Authentic Human Engagement via Relational Tables

Unlike platforms that manipulate arbitrary integer fields, Liread enforces authentic community engagement:

- **Relational Uniqueness:** `post_upvotes` enforces `UNIQUE(post_id, user_id)`, strictly guaranteeing that one user can only like a passage once.
- **Nested Relation Counting:** Client queries `post_upvotes(user_id)` directly to compute true real-time upvotes without trusting inflated demo integers.
- **Zero Counter Leakage:** When no users have upvoted a post, it strictly displays **0 / "No likes yet"**.

Database Triggers & Row-Level Security

Idempotent RLS policies and automated PL/pgSQL event triggers

Automated PL/pgSQL Triggers

When a user finishes a timer session, `on_reading_session_created` recalculates continuous daily streaks and promotes the reader's rank tier atomically on the database engine.

```
EXECUTE FUNCTION handle_reading_session_inserted();
```

Strict Row-Level Security (RLS)

Every table has RLS enabled. Users can only edit or delete their own bookshelf items and marginalia, while public feeds and profiles remain safely accessible via authenticated and guest read policies.

```
WITH CHECK (auth.uid() = user_id);
```

Horizon 1: AI Contextual Marginalia

Demystifying classic literature inline without breaking reading flow

Context-Aware Literary Companion & Semantic Search

Q1 2027

Reading dense classic philosophy (Marcus Aurelius, Spinoza, Kant) can be intimidating. Horizon 1 introduces on-demand contextual prompts directly inside the reading canvas:

- **Zero-Spoiler Concept Explanations:** Tap any complex passage for a concise 2-sentence historical and philosophical background briefing.
- **Semantic Search Across Saved Marginalia:** Retrieve quotes by concept (e.g. *"passages about dealing with anxiety and mortality"*) using vector embeddings.
- **Archaic Lexicon Translator:** Instant definitions of Victorian and ancient idioms in natural modern prose.

Horizon 2: Asynchronous Micro-Clubs

Shared chapter margins and voice annotations with spoiler protection gates

Synchronous Pacing with Asynchronous Freedom

Q2 2027

Traditional book clubs struggle with scheduling meetings. Horizon 2 brings collaborative margins to Liread:

- **Spoiler-Safe Chapter Gates:** Friends' reflections on Chapter 4 only unlock once you finish reading Chapter 4 yourself.
- **Audio Voice Marginalia:** Attach 30-second audio thoughts directly to memorable book paragraphs.
- **Classroom & Reading Circle Groups:** Dedicated private sanctuaries for university courses and small friend circles.

Horizon 3: Native Mobile & E-Ink Hardware Sync

Bridging physical E-readers and mobile devices into the Liread sanctuary

Hardware Integration & High-Contrast E-Ink Optimization

Q3 2027

Empowering readers who prefer electronic ink displays or mobile lock-screen widgets:

- **Kindle & Kobo Integration:** Bi-directional synchronization of annotations and reading progress directly into Liread.
- **High-Contrast E-Ink Mode:** Zero-animation, monochrome high-DPI stylesheet optimized for Onyx Boox and Kindle web browsers.
- **Lock-Screen Habit Widgets:** iOS Live Activities and Android glanceable widgets displaying today's focus progress.

THE LIREAD THESIS

"Reading is not dying. It is simply waiting for a *calm sanctuary* to call home."

By marrying distraction-free audio engineering, active in-reader notification and app blocking, zero-friction library access, and honest human accountability, Liread empowers readers worldwide to reclaim their attention and build lifelong compounding wisdom.

Let's Make the World Read Again.

Experience Liread live right now on your phone or laptop.

LIVE APPLICATION

`liread.vercel.app`

SOURCE REPOSITORY

`mulukentsegaye/liread-web`

Thank you! Questions & Live Demonstration